Haoran (Alex) Yu

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EDUCATION

Boston University Metropolitan College **Boston, USA**
Master of Science in Applied Data Analytics *Expected Entry: January 2026*

University of Toronto, St. George Campus **Toronto, Canada**
Major in Mathematics (Minors in Statistics and Economics), HBSc *September 2021 – July 2025*

- **Courses Covering Topics in:** Linear Algebra, Advanced Calculus, Financial Economics and Corporate Finance, Mathematical Finance, Micro and Macro Economics, Game Theory, Group Theory, Graph Theory, Real and Complex Analysis, Mathematical Modelling, Nonlinear Optimization, Probability and Statistics, Data Analysis, Machine Learning, Statistical Survey Sampling

Projects

Credit Risk Analysis of Enbridge Inc. ***Toronto, Canada***
Independent Project *March 2025 – April 2025*

- Based on the Merton Model, used the 15-year Canadian government bond yield as the risk-free rate and annualized volatility of Enbridge's stock price (2010–2025) to iteratively solve for asset volatility.
- Estimated the firm's distance to default and probability of default over the next 15 years.
- Built a two-step transition probability matrix assuming a 50% recovery rate and assessed credit risk based on the bond spread between Enbridge and U.S. Treasury bonds.
- Created a one-page A4 poster presenting the project's background, methodology, key findings, and conclusions.

Loan Default Risk Prediction Using Machine Learning ***Toronto, Canada***
Independent Project *August 2024 – December 2024*

- Used the Lending Club dataset and performed data cleaning, missing value handling, and feature encoding using pandas.
- Explored variable distributions and correlations using Matplotlib and pandas to identify key predictors of loan defaults.
- Built and tuned a Random Forest model using scikit-learn with cross-validation to optimize performance.
- Evaluated model performance through confusion matrix, classification report, and ROC/AUC curve analysis.
- Entire pipeline developed and visualized in Jupyter Notebook.

Vehicle Insurance Premium Modeling ***Toronto, Canada***
Team Leader and Group Project *July 2024 – August 2024*

- Led team management and project planning, and independently developed core algorithms in R.
- Used R to construct linear regression models based on real-world Spanish auto insurance data with data cleaning and transformations.
- Applied Box-Cox, square root, logarithmic transformations, and ridge regression for model optimization.
- Conducted statistical tests including F-test, partial F-test, Cook's Distance, and R-squared evaluations to ensure model reliability.
- Authored a complete report in LaTeX detailing methodology, results, and other analysis.

Internship Experience

Shaanxi Jintai Hengye Real Estate Co., Ltd. ***Shaanxi, China***
Real Estate Investment Expansion Intern *June 2023 – September 2023*

- Assisted in collecting and systematically organizing land auction data as well as legal and financial documents related to acquisitions
- Participated throughout the early stages of project development, including market research, financial modeling, and risk assessment
- Contributed to the drafting of feasibility study reports under the guidance of senior analysts, gaining hands-on experience in core processes and professional methods of commercial real estate project analysis

Programming & Tools:

Python and libraries (NumPy, pandas, Matplotlib, scikit-learn, SciPy), DEAP (Distributed Evolutionary Algorithms in Python), R and tidyverse, Git, LaTeX, SQL